

Numerical Analysis of Heat Sink Design for Thermal Management of Electronics

A PROJECT REPORT

Submitted by

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Under the guidance of

**Dr. NITESH MONDAL
Assistant Professor**

Department of Mechanical Engineering

In partial fulfilment for the award of the degree

Of

**BACHELOR OF TECHNOLOGY
IN
MECHANICAL ENGINEERING**



Ghani Khan Choudhury Institute of Engineering & Technology

गनी खान चौधरी इंजीनियरिंग और प्रौद्योगिकी संस्थान

A Centrally Funded Technical Institute(CFTI)

under Ministry of Education Govt of India

(शिक्षा मंत्रालय, भारत सरकार के तहत सीएफटीआई)

JULY 2026

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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RECOMMENDATION

The report titled '**NUMERICAL ANALYSIS OF HEAT SINK DESIGN FOR THERMAL MANAGEMENT OF ELECTRONICS**' carried out by **ANUPAM DAS, ARNAB PRAMANIK, AUGNIK DAS, SRIJIT GOSWAMI, RAHUL GHOSH, AISHIK DUTTA** under my supervision is recommended as PROJECT REPORT in *partial fulfilment for the award of the degree of **Bachelor of Technology in Mechanical Engineering.***

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BONAFIDE CERTIFICATE

Certified that this project report “**NUMERICAL ANALYSIS OF HEAT SINK DESIGN FOR THERMAL MANAGEMENT OF ELECTRONICS**” is the Bonafide work of **ANUPAM DAS, ARNAB PRAMANIK, AUGNIK DAS, SRIJIT GOSWAMI, RAHUL GHOSH, AISHIK DUTTA** and is accepted as PROJECT REPORT for *partial fulfilment for the award of the degree of* **Bachelor of Technology in Mechanical Engineering**.

----- Examiners	----- Dr. NITESH MONDAL Assistant Professor
----- Dr. NITESH MONDAL Head of the Department DEPARTMENT OF MECHANICAL ENGINEERING	

ABSTRACT

Due to the increasing trend of power density in modern electronics devices, effective thermal management becomes a necessary requirement for providing reliability, performance, and life of the device. There are several cooling methods among which air-cooled plate-fin heat sinks have been extensively used due to their simplicity, low manufacturing cost, and easy maintenance. The traditional straight fin heat sink is constrained by the thermal boundary layer effect that decreases the heat transfer efficiency. In order to overcome the problem, the current research suggests the use of SPF heat sink for cooling electronic chips.

To begin with, the benchmark model of a plate-fin heat sink proposed by Singh et al. has been simulated and verified with ANSYS Fluent 2026 R1 Student Version. The numerical modelling has been carried out using SolidWorks, ANSYS Space Claim, and ANSYS Meshing and the results have been verified with the benchmark data. It has been found that there is good agreement between simulation and experimental values with an average error of around **2.03%**.

The above-validated benchmark model has been used as a base to design a new SPF heat sink by adding staggered circular perforations to the geometry without altering any other parameter like dimension or operation parameters. The numerical simulations have been conducted under six different operation parameters by changing inlet air velocity, inlet air temperature, and volumetric heat generation rate.

As indicated above, the results from the simulation showed that the proposed SPF heat sink could provide reduced temperatures compared to the traditional straight-fin heat sink. The temperature reduction varied between **3.430 K** and **8.244 K** with an average temperature reduction of **5.966 K** representing an improvement of **1.90%** in terms of thermal properties. This could be attributed to better airflow distribution, more fluid mixing, and disturbance of thermal boundary layer due to the staggered perforation of the heat sink.

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NOMENCLATURE

Symbol	Description	Unit
V	Air velocity	m/s
T	Temperature	K
T_{air}	Inlet air temperature	$^{\circ}C$
T_{chip}	Average chip temperature	K
q_{gen}	Volumetric heat generation	W/m^3
k	Thermal conductivity	$W/m \cdot K$
C_p	Specific heat capacity	$J/kg \cdot K$
ρ	Density	kg/m^3
h	Convective heat transfer coefficient	$W/m^2 \cdot K$
Q	Heat transfer rate	W
P	Pressure	Pa
Re	Reynolds number	—
μ	Dynamic viscosity	$Pa \cdot s$
ε	Turbulent dissipation rate	m^2/s^3
k (Turbulence)	Turbulent kinetic energy	m^2/s^2

ABBREVIATIONS

Abbreviation	Full Form
CFD	Computational Fluid Dynamics
SPF	Staggered Perforated Fin
CAD	Computer-Aided Design
CPU	Central Processing Unit
GPU	Graphics Processing Unit
ECU	Electronic Control Unit
IGBT	Insulated Gate Bipolar Transistor
MOSFET	Metal–Oxide–Semiconductor Field-Effect Transistor
ANSYS	Analysis System
RNG	Renormalization Group
SIMPLE	Semi-Implicit Method for Pressure-Linked Equations
VLSI	Very Large-Scale Integration

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CHAPTER-1

1.1 Overview

Fast growth of electronic technology made it possible to create small, light-weight and highly efficient electronic equipment. The constant increase in the density of transistors and capacity of processors has caused an increase in heat generation in the electronics' chips. The excess heat leads to deterioration in the effectiveness, reliability, and longevity of electronic devices. Therefore, thermal management has become one of the key design issues in today's electronics.

There are many thermal management techniques which can be employed. However, air-cooling heat sinks are extensively used due to their easy design, low manufacturing price, ease of maintenance, and reliability. Several factors determine the performance of a heat sink, including fin geometry and fin spacing, material properties, airflow velocity and heat generation rate. Optimization of these factors will help to achieve better performance.

Computer Fluid Dynamics (CFD) is considered to be an effective numerical tool for analysis of heat transfer and flow process in electronics. CFD allows designers to calculate such parameters as temperature profile, airflow characteristics and pressure drops without any prototype production. This helps to decrease development costs and time significantly.

This work is derived from the benchmark research paper entitled “**Optimization and Numerical Simulation of Heat Sink for Cooling Electronic Chip**”. The benchmark heat sink model has been replicated through ANSYS Fluent to verify the numerical method in similar operational conditions. Upon successful verification, a new design of **Staggered Perforated Fin (SPF)** heat sink has been devised by creating uniformly distributed circular perforations in a staggered arrangement while keeping the original benchmark dimensions. This modification would facilitate airflow interaction, disruption of thermal boundary layer, increase convection and lower thermal resistance.

The contents of the current thesis include literature review, benchmark model design, CAD modelling, computational domain creation, mesh generation, numerical simulation, benchmark validation and comparative performance analysis of the proposed SPF heat sink. Thermal behaviour of both the conventional and modified heat sink models has been studied under same boundary conditions in ANSYS Fluent.

1.2 Introduction

With the fast development of modern electronics such as advanced microprocessors, power electronics converters, telecommunication equipment, and control units, a lot of heat is generated in the chips. As the electronic components keep getting smaller and more powerful, the heat flux increases a lot, making thermal management a very important issue in the design of electronic systems [1]-[6].

Excessively high operating temperatures have a detrimental effect on the functionality, reliability, and lifetime of electronic components. Excessively hot chips can cause stress to the components, increased leakage currents, decreased operational efficiency, accelerated aging of materials used, and ultimately the failure of the component [3], [4], [6]. This is why an efficient cooling technique that would keep the operating temperature under control has become a necessity.

Numerous cooling methodologies have been suggested to solve this problem such as natural air cooling, forced air cooling, liquid cooling, heat pipes, vapor chamber, PCM, thermoelectric cooler, and nanofluid based cooling systems [5], [17] – [20]. Of these cooling technologies, the one that is most commonly used is the air-cooled heat sink due to its simple structure, economical, reliable, easy fabrication, and maintenance free properties [6], [18], [19].

There are various design factors that determine the efficiency of an air-cooled heat sink, which include the geometry of the fins, spacing between fins, height of the fins, thickness of the fins, size of the base plate, materials and the flow of the air [7], [8], [20]. Therefore, considerable research efforts have been made to optimize these design factors to achieve efficient heat transfer with minimum pressure loss and use of material.

Recent developments in computational fluid dynamics have facilitated the investigation of the complex heat transfer and flow behaviours in the heat sink configurations without entirely depending on the costly experimental studies. The CFD study offers comprehensive details about the temperature profile, velocity distribution, pressure variation, thermal resistance and airflow behaviour [23] – [27].

In the recently published literature, the benchmark study performed by Singh, Singh, and Trivedi (2025) provides a systematic analysis for optimizing an air-cooled plate-fin heat sink using the CFD approach for electronic chip cooling [9]. The benchmark study proved to be successful in proving that geometric parameter optimization can result in reduced average chip temperatures. The benchmark study with the comprehensive numerical technique and clear definition of operating conditions is thus the suitable model to conduct further research on.

Although the benchmark model shows great thermal improvement, it is based on the use of conventional solid fins. In low-to-medium Reynolds numbers, the conventional solid fin designs lead to the development of thermal and hydrodynamic boundary layers on the surface of the fins. With the increase in thickness of the boundary layer, the heat transfer coefficient drops making further improvement of cooling impossible [10] – [14], [28], [29]. Also, conventional solid fins limit the flow interaction inside the fins and use comparatively more material than necessary, making them heavier without improved heat transfer properties.

Many authors have shown that incorporation of perforations in fin design can efficiently disrupt the thermal boundary layer, increase mixing of airflow, produce vortex flows, and thus enhance convective heat transfer along with saving materials [21], [31] – [37]. The staggered perforation design further helps in disrupting the boundary layer constantly and uniformly distributing the airflow, and thus offers better thermal performance than conventional solid fins [33] – [39].

Based on these observations, the current research utilizes the benchmark model proposed by Singh et al. [9] and reproduces and verifies it using ANSYS Fluent at the same operational conditions. On successful verification of the benchmark model, a new design of Staggered Perforated Fin (SPF) heat sink is created by creating holes of circular cross-section in the fin arrangement without any change in the benchmark dimension. Thermal performances of both designs are analysed under same boundary conditions.

The result of this study will lead to an efficient passive cooling technique that can be used for electronic chip cooling, thereby contributing to the development of efficient heat sinks for future electronic devices [9], [31] – [39].

1.3 Problem Statement

The continuous increase in power density of modern electronic devices has made thermal management a major engineering challenge. Conventional plate-fin heat sinks are widely used for electronic cooling; however, their performance is often limited by thermal boundary layer formation and restricted airflow between fins, resulting in higher chip temperatures and reduced cooling efficiency. Although several studies have proposed different fin geometries, limited work has been carried out on staggered perforated fin (SPF) configurations using a validated benchmark model. Therefore, there is a need to investigate whether an SPF heat sink can improve thermal performance while maintaining the same operating conditions as the benchmark model.

1.4 Objectives

The primary goals of the current study are as follows:

- Reproduction of the benchmark geometry of the heat sink by utilizing ANSYS SpaceClaim.
- Generation of the computational domain along with mesh generation.
- Validation of the benchmark model by comparing simulation results with benchmark results.
- Designing of the Staggered Perforated Fin (SPF) heat sink by using the benchmark model.
- Analysis of the thermal performance of both benchmark and SPF heat sinks by using ANSYS Fluent.
- Comparison of temperature profiles and airflow characteristics between benchmark and SPF heat sinks.

1.5 Scope of the Current Research

In the current research, attention is paid to the numerical analysis of an air-cooled plate fin heat sink for electronic chip cooling. Firstly, the benchmark model is designed and validated in ANSYS Fluent. Subsequently, the Staggered Perforated Fin (SPF) model is created by perforating the benchmark model with circular holes but keeping other parameters unchanged. Comparison between the benchmark and SPF models is made through various analyses such as chip temperature, temperature profiles, air flow distribution, and efficiency.

1.6 Research Methodology

The research methodology used in the current study includes the following stages:

- a) Identification of the benchmarking research paper.
- b) Building the benchmarking CAD model using ANSYS SpaceClaim.
- c) Model domain and mesh generation.
- d) Material and boundary condition assignment.
- e) CFD simulation and benchmark model verification using ANSYS Fluent.

- f) Design of the SPF heat sink.
- g) Numerical analysis of the SPF model under the same conditions.
- h) Benchmark and SPF model comparison.
- i) Analysis of temperature distribution and thermal performance.
- j) Conclusion and recommendations for further research.

1.7 Structure of the Thesis

The thesis is divided into five chapters.

Chapter 1 includes introduction, problem statement, objective of work, scope of work, methodology and structure of the thesis.

Chapter 2 includes literature review on the topic of e-cooling, heat sinks, CFD simulation, benchmarking studies, behaviour of thermal boundary layer and perforated fins arrangements. Research gap is also highlighted in this chapter.

Chapter 3 includes development of benchmark model, CAD modelling, mesh generation, numerical simulation and validation of the results with the benchmark paper.

Chapter 4 includes development of the novel Staggered Perforated Fin (SPF) heat sink, numerical simulation, analysis of the results and comparison of the results with validated benchmark model.

Chapter 5 includes major findings of the thesis, conclusions drawn, contributions of the work and future recommendations.

CHAPTER-2

LITERATURE REVIEW

2.1 Introduction

A thorough review of related literature is necessary to identify the existing trends in electronic cooling systems and possible ways to enhance their performance. In recent decades, many researchers have introduced different thermal management techniques to dissipate the heat released by high power electronic components. One of the most popular techniques in terms of its simplicity, reliability, and low cost is air-cooled heat sinks [1] – [6].

Many researchers have considered heat sink optimization problems, changing fins shape, spacing, materials, and airflow parameters. With the development of CFD technology, it became convenient to numerically simulate the temperature fields, airflow properties, and pressure drops before manufacturing heat sinks [7] – [10].

The latest researches paid much attention to the perforated fins and other fins modifications that could increase the performance of heat sinks by improving the convective heat transfer rate and decreasing the thermal resistance [11] – [15]. The selected benchmark in the present work represents a reliable numerical technique for heat sink performance estimation under forced convection [9].

Literature related to electronic cooling methods, standard heat sink geometries, thermal analysis using CFD, benchmarking, and perforated fin geometries has been reviewed in this chapter. This literature review would serve the purpose of identifying the existing research gap and forming the basis of the Staggered Perforated Fin (SPF) heat sink studied in this work.

2.2 Electronic Device Cooling

Because of the increasing density of energy in modern electronic devices, it became necessary to implement proper thermal management in their design. During the working process, electronic elements such as microprocessors, power modules, and integrated circuits produce a substantial amount of heat. In case there is no effective dissipation of this heat, the temperature of the device increases resulting in its lower efficiency, lower reliability, thermal stresses, and shortened life span [1] – [4].

In order to maintain the safe temperature level in electronic devices, different cooling methods are used, namely natural convection, forced air cooling, liquid cooling, heat pipes, vapor chambers, and microchannel cooling systems [5] – [8]. Among all the cooling methods mentioned above, air cooling with the help of heat sinks is the most commonly applied one owing to its simplicity, low manufacturing costs, easy maintenance and efficient performance [6], [9]

There are a number of aspects influencing the efficiency of the electronic cooling system such as heat sink shape, fins distribution, material characteristics, airflow conditions, and heat generation rate. Therefore, continuous researches are carried out aimed at optimizing heat sinks in order to increase heat transfer rate and minimize pressure drop and material consumption [9] – [12].

This research is based on the research conducted by Singh et al. [9], and it examines the efficiency of an improved SPF heat sink compared to existing plate-fin heat sinks under the same operational parameters.

2.3 Conventional Heat Sink Technologies

Heat sinks are passive devices which effectively cool down electronic devices by increasing the surface area through which heat is transferred out. Several kinds of heat sinks, depending on their applications and need for cooling have been made, among which include plate-fin, pin-fin, folded-fin and micro-channel heat sinks [6] – [9].

Of all these kinds of heat sinks, the plate-fin heat sink is the most commonly used owing to its simple structure, ease of manufacturing, lower costs and efficient thermal behaviour when using forced air cooling [9],[17]. Thermal performance of heat sinks depends on many factors such as fin geometry, fin spacing, fin height, material used, and airflow velocity. It has been shown that optimizing these factors leads to increase in heat dissipation and decrease in thermal resistance [10] - [13].

Although solid-fin heat sinks perform well enough, there is the possibility of having less heat dissipation due to thermal boundary layers formation on the fins surface. Therefore, several designs of fins have been made by researchers to increase airflow and improve heat dissipation capacity [14]-[16].

2.4 Computational Fluid Dynamics (CFD) in Heat Sink Analysis

Computational Fluid Dynamics (CFD) is a numerical method applied for analysing fluid flow and heat transfer phenomena in engineering systems. For the purposes of electronic cooling, CFD provides a means of predicting temperature distribution, air flow characteristics, pressure losses, and thermal performance without resorting to expensive experimental tests [18] – [21].

The ANSYS Fluent CFD program is the most commonly used software tool for performing thermal analyses of heat sinks. With the help of this program, it is possible to investigate different scenarios, optimize the geometry of the heat sink and examine the impact of design changes on the cooling efficiency [22] – [25]. Compared to experimental techniques, CFD takes less time and money to develop and offers detailed information about heat transfer processes [23], [24].

In the current research, the ANSYS Fluent program is applied to create the benchmark model and investigate the performance of the suggested Staggered Perforated Fin (SPF) heat sink under the same boundary conditions [9].

2.5 Benchmark Study

The benchmarking study chosen for the current project provides an accurate numerical basis for the analysis of the thermal efficiency of plate-fin heat sink for the cooling of electronic chips. The study examined the effects of geometry and operational parameters on temperature distribution via ANSYS Fluent. The results from the numerical analysis were shown to be valid and in agreement with the existing data, proving the validity of the applied CFD approach [9].

In the current work, the benchmark geometry, material properties, boundary conditions, and solver settings will be duplicated to validate the numerical model. Once the model has been validated, it will serve as a reference point for comparison of the proposed Staggered Perforated Fin (SPF) heat sink [9].

2.6 Thermal Boundary Layer Formation

As the flow of air passes over the heat sink, a thermal boundary layer is formed because of the temperature differences between the surface of the fins and the air. The formation of the thermal boundary layer is affected by the airflow; as the air moves, the thickness of the thermal boundary layer increases. Thus, the thermal boundary layer limits the heat dissipation capacity of the heat sink [10] – [14].

Formation of the thermal boundary layer affects the process of heat dissipation. Hence, minimizing or breaking the thermal boundary layer is one effective method of improving thermal performance and heat dissipation from electronic parts [14] – [16].

2.7 Boundary Layer Disruption Technique

To increase the thermal effectiveness of heat sinks, a variety of approaches to disturb the thermal boundary layer have been developed. Such approaches can include interrupted fins, wavy fins, vortex generators, offset fins, and perforated fins. This increases the mixing of the flow, increases turbulence, and increases the interaction of fresh air with the heated fin surfaces, resulting in improved convective heat transfer coefficient [15] – [18].

One of such approaches is using perforated fins, which provide an increase in flow through the fin, decrease the boundary layer thickness, and enhance cooling efficiency with relatively low additional pressure loss [19] – [21]. These results give rise to the idea of analysing various modifications of perforated fins for cooling of electronics.

2.8 Staggered Perforated Fin (SPF) Heat Sink

Staggered perforated heat sinks have proved themselves to be a suitable way for passive cooling systems to increase the efficiency of heat transfer in electronics. The advantage of having perforations provides for better air flow distribution and disturbing the thermal boundary layer along with an increased rate of heat transfer by convection [22]– [26].

In the current investigation, the novel design called the Staggered Perforated Fin (SPF) is introduced where circular perforations are provided in a staggered pattern while keeping the same dimensions of the benchmark heat sink. The performance analysis of SPF heat sink is carried out using ANSYS Fluent and compared with the conventional benchmark heat sink.

2.9 Research Gap

Based on the literature review, a lot of research work has been done related to the conventional plate-fin heat sinks, fin geometry optimization, and thermal analysis of heat sinks using CFD. Even though perforated fins have proven to perform well regarding heat transfer performance, very few works have been performed on the Staggered Perforated Fin (SPF) heat sinks under the same geometry and operating conditions. Also, there are very few comparative numerical validations as well as performance analysis of the heat sinks [9]-[15], [22] - [26].

Hence, in this study, the objective is to develop and validate a benchmark heat sink model, and then evaluate the thermal performance of the Staggered Perforated Fin (SPF) heat sink using ANSYS Fluent.

2.10 Chapter Summary

In this chapter, the literature on electronic cooling, traditional heat sink techniques, CFD analysis, benchmark analysis, behaviour of thermal boundary layer, and perforated fins has been studied. In this review, the research gap was found out, and the possibility of using staggered perforated fins was noted. With these results in view, benchmark model has been selected, and a Staggered Perforated Fin (SPF) heat sink has been recommended for numerical analysis in upcoming chapters [9] – [26].

CHAPTER-3

BENCHMARK MODEL DEVELOPMENT AND NUMERICAL METHODOLOGY

3.1 Introduction

This chapter explains the design and numerical verification of the benchmark heat sink model used in the current study. Prior to proposing the Staggered Perforated Fin (SPF) structure, it is necessary to ensure that the CFD model correctly simulates the thermal performance observed in the benchmark study. Thus, the benchmark model proposed by Singh et al. [9] was reproduced using ANSYS Space Claim and the numerical simulations were conducted in ANSYS Fluent with similar operational conditions.

The first step in the modelling procedure was developing the geometry of the electronic chip, the heat sink and the air enclosure. Suitable material properties, boundary conditions and other solver settings were defined based on the benchmark study [9]. An appropriate computational mesh was designed in order to provide accurate simulation of fluid flow and heat transfer parameters at a relatively low computational cost.

Upon completing the numerical simulation process, the distribution of temperature and the average chip temperature were obtained. They were compared with the benchmark data from Singh et al. [9]. The good correlation of the present numerical results with the benchmark values verifies the accuracy of the designed CFD model.

The benchmark model, which has been verified, serves as the foundation for developing and analysing the SPF heat sink proposed in the following chapter. It guarantees that the thermal performance comparison study will be reliable by verifying the numerical approach prior to any changes to its geometry.

3.2 Benchmark Heat Sink Geometry

The benchmark geometry for the heat sink geometry in the current investigation was adopted from Singh et al. [9]. The benchmark geometry was selected as the reference geometry in order to verify the numerical procedure prior to analyzing the current proposed Staggered Perforated Fin (SPF) heat sink. All dimensional information for the benchmark geometry is maintained in accordance with the published results [9].

The benchmark geometry comprises an electronic chip, aluminum base plate and five rectangular fins that are evenly distributed on the base surface. The electronic chip has the dimensions of **10 mm × 10 mm × 1 mm** and is mounted below the heat sink base. All fins have the dimensions of **10 mm × 10 mm × 0.8 mm** with the fin spacing of **1.5 mm**, ensuring that air can flow between the adjacent fins [9].

The benchmark geometry was replicated using the SolidWorks package by referring to the dimensions reported in the reference study. The complete CAD geometry was validated in

terms of dimensional accuracy and then exported for subsequent processing and numerical analysis in ANSYS Fluent. The benchmark geometry of the heat sink for the current study is presented in Figure 3.1.

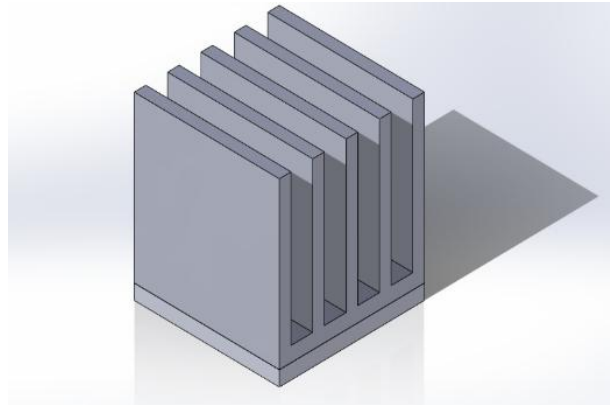


Figure 3.1: Three-dimensional CAD model of the benchmark plate-fin heat sink developed in SolidWorks.

Component	Dimension
Chip	$10 \times 10 \times 1$ mm
Fin height	10 mm
Fin length	10 mm
Fin thickness	0.8 mm
Number of fins	5
Fin spacing	1.5 mm

Table 3.1 Geometrical dimensions of the benchmark heat sink

3.3 CAD Model Development

A 3D CAD model of the benchmark heat sink was first created with SolidWorks 2023 based on the geometric parameters of the benchmark study by Singh et al. [9]. This geometry consists of an electronic chip, a base plate, and five straight rectangular fins that are placed symmetrically on the base surface. Geometry modelling was carried out ensuring all the benchmark geometric dimensions to match the benchmark reference geometry.

After the completion of the CAD model, it was exported in STEP format (.stp) and imported into ANSYS SpaceClaim for further preprocessing. During this process, the geometry was checked for any possible gaps, overlapping geometry, or invalid edges that

would affect meshing. All necessary geometry checks were performed to ensure that the geometry model is ready for CFD analysis.

Geometry preparation for creating a computational domain involved setting the required fluid domain surrounding the geometry model for airflow simulation. The benchmark heat sink geometry model imported in ANSYS SpaceClaim is presented in Fig. 3.2.

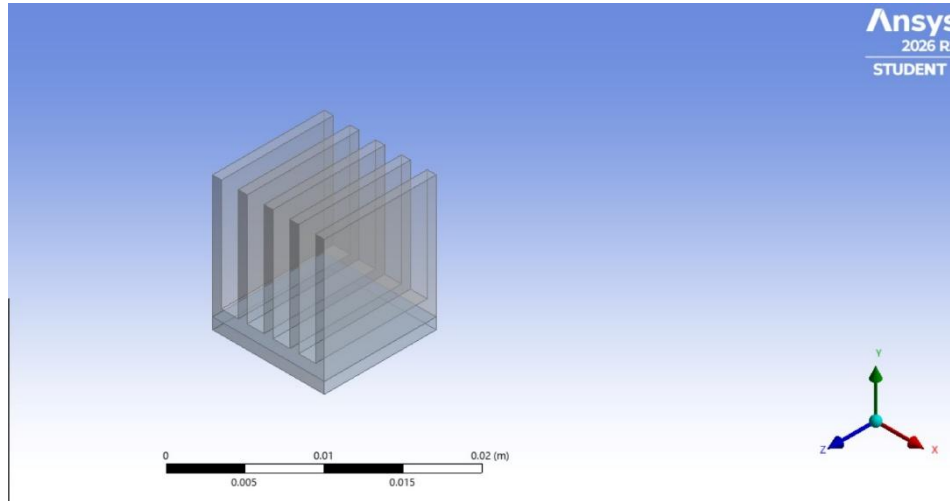


Figure 3.2: Benchmark heat sink geometry imported into ANSYS SpaceClaim for CFD preprocessing.

3.4 Computational Domain

In order to perform the forced convection cooling analysis of the heat sink benchmark problem, the computational domain was developed around the solid model using the Enclosure feature of ANSYS SpaceClaim. The computational domain represents the surroundings in which the air flows through the heat sink during the numerical simulation process. The development of the computational domain is important for obtaining accurate results in terms of the fluid flow and heat transfer without the effect of boundary conditions [9].

The dimensions of the computational domain were based on the benchmark problem such that the same numerical conditions could be obtained. An inlet zone of **15 mm** was established before the heat sink to provide a properly developed flow before entering the fins, while an outlet zone of **50 mm** was established after the fins to avoid the effect of any reverse flow due to the presence of outlet walls. Also, enough clearance was provided in the vertical and lateral directions to avoid the effect of wall boundaries [9].

Once the enclosure was created, the air region surrounding it was considered as the fluid domain, while the electronic chip and heat sink were considered as the solid domains. This approach helped simulate conjugate heat transfer in between the solid components and the surrounding air. Once the entire domain was generated, it was moved to the ANSYS Meshing for further meshing and numerical computations.

Select one or more bodies to create an enclosure around. Click the checkmark tool guide to complete the enclosure.

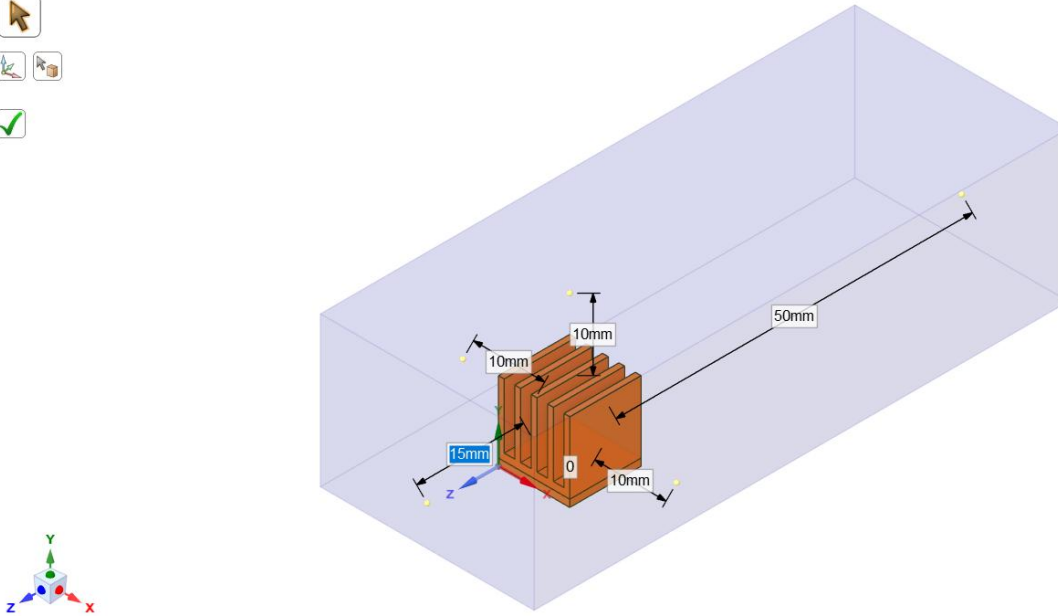


Figure 3.3: Computational domain (air enclosure) created around the benchmark heat sink using ANSYS SpaceClaim.

3.5 Mesh Generation

The benchmark heat sink geometry obtained from ANSYS SpaceClaim was exported to ANSYS Meshing to create the computational grid. Mesh creation can be termed as one of the most crucial steps in Computational Fluid Dynamics (CFD) since the governing equations of mass, momentum, and energy conservation are solved over a discrete set of control volumes [9]. In addition, the quality of the mesh will directly impact the accuracy, stability, and convergence behaviour of the simulation.

In this study, an unstructured tetrahedral mesh was created for the whole computational domain which consisted of the electronic chip, heat sink, and the air enclosure. Tetrahedral elements were chosen for creating the mesh due to the complexity of the problem in three dimensions and also due to good quality of mesh that can be achieved by such type of elements. A global element size of 1.5 mm was considered for the whole computational domain while a body sizing was applied on the electronic chip and the heat sink where the temperature and velocity gradient are very large. The mesh was thoroughly checked using various mesh quality criteria before importing it to ANSYS Fluent.

The generated computational mesh is shown in **Figure 3.4**, while the mesh quality parameters are presented in **Figures 3.5–3.8**.

3.5.1 Generated Computational Mesh

Figure 3.4 shows the mesh generated for the benchmark heat sink and the air cavity that surrounds the heat sink. In this figure, the mesh has tetrahedral cells in both the fluid and solid domains. Finer mesh was used in the regions close to the heat sink fins and the electronic chip to predict better heat transfer and fluid flow while coarser mesh was employed in the regions far from these areas to minimize computation costs.

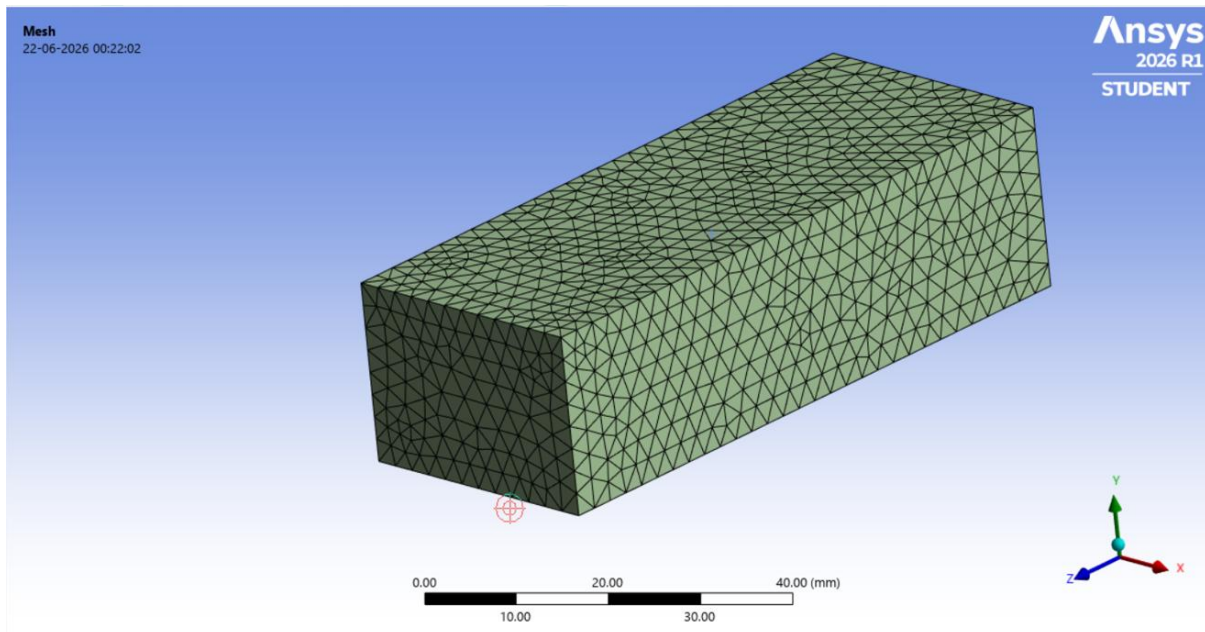


Figure 3.4: Generated computational mesh of the benchmark heat sink.

3.5.2 Mesh Quality Analysis

Once the mesh generation process is completed, then the quality of the computational mesh was analyzed based on the quality criteria provided by ANSYS Meshing. Some of the common quality criteria include Skewness, Orthogonal Quality, Element Quality, and Aspect Ratio. The results have been compared with the recommended range of ANSYS Fluent for reliable simulation.

3.5.2.1 Skewness

The Skewness ratio is used to measure the difference between an element and the ideal element shape. Low skewness values suggest good mesh quality and increased numerical stability. The maximum skewness value in the present mesh is **0.799**, which lies under the acceptable limit of **0.85** for CFD simulations. Thus, the mesh generated is suitable for numerical analysis.

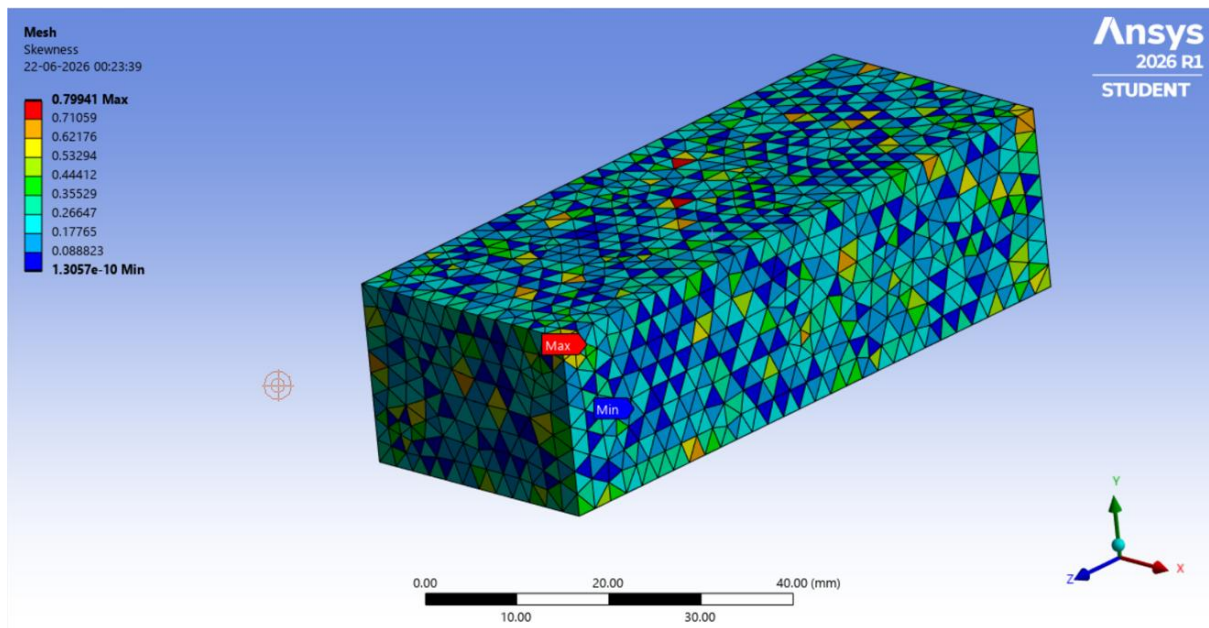


Figure 3.5: Skewness distribution of the computational mesh.

3.5.2.2 Orthogonal Quality

Orthogonal quality measures the relationship between the neighbouring mesh entities. Higher orthogonal quality values reflect high-quality meshes and low discretization errors. The lowest orthogonal quality value calculated is **0.2006**, which is above the minimum recommended value of **0.15** for ANSYS Fluent. This ensures that the created mesh has sufficient orthogonality for reliable computations

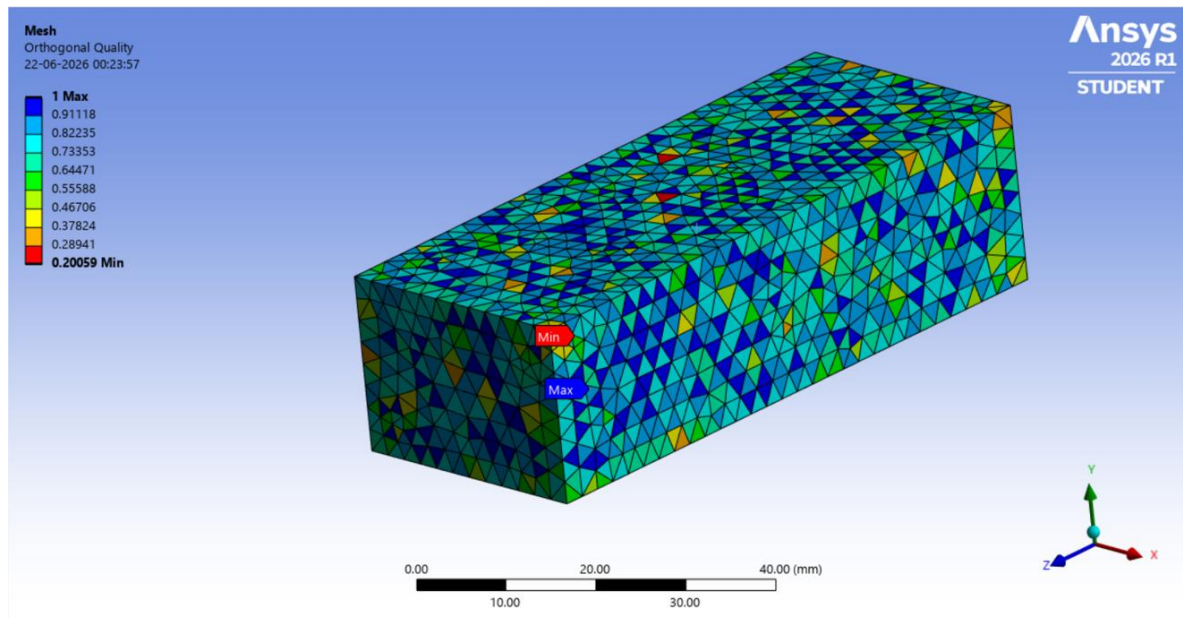


Figure 3.6: Orthogonal quality distribution of the computational mesh

3.5.2.3 Element Quality

The quality of an element denotes the geometrical quality of individual mesh elements. The closer the values are to one, the better is the quality of the element and greater the accuracy of the results. The minimum element quality achieved in the current study is **0.2685**, which is quite satisfactory in comparison to the minimum threshold required for CFD analysis.

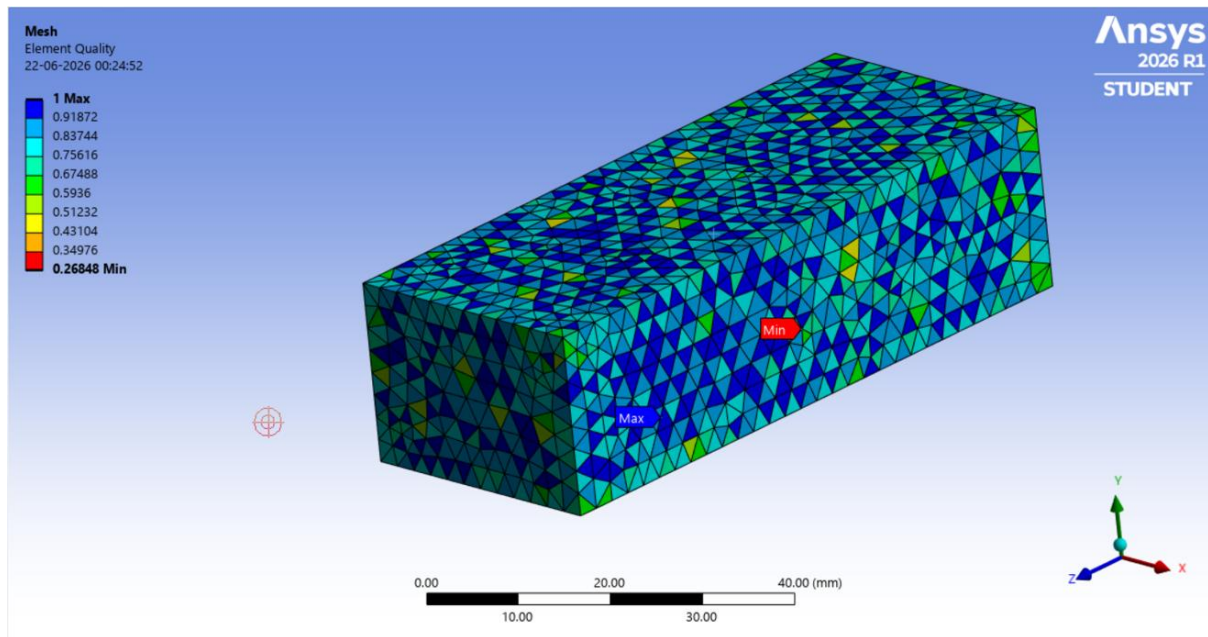


Figure 3.7: Element quality distribution of the computational mesh.

3.5.2.4 Aspect Ratio

The aspect ratio can be described as the ratio of the longest side to the shortest side in any mesh element. Smaller aspect ratios enhance numerical stability and help to minimize interpolation errors. The highest aspect ratio found in the current mesh is **7.7755**, and this is within the range of less than 10 that is recommended. Thus, the mesh is acceptable.

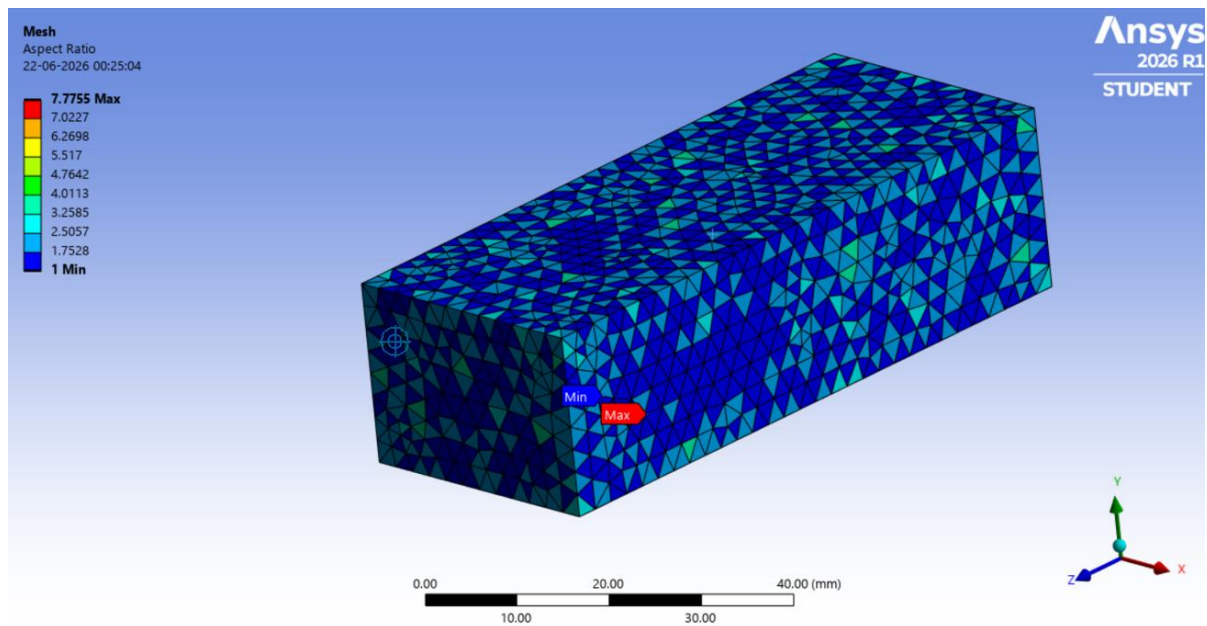


Figure 3.8: Aspect ratio distribution of the computational mesh.

3.5.3 Summary of Mesh Quality

The quality of the mesh obtained through ANSYS Meshing reveals that the computational mesh generated fulfils the necessary requirements for performing ANSYS Fluent analysis. All the values of skewness, orthogonal quality, element quality, and aspect ratio fall within acceptable ranges, providing accurate results related to fluid dynamics and thermal conductivity. Consequently, the mesh generated was used for performing benchmark and numerical analysis.

Mesh Parameter	Obtained Value	Recommended Limit	Status
Global Element Size	1.5 mm	—	Acceptable
Maximum Skewness	0.799	< 0.85	✓
Minimum Orthogonal Quality	0.2006	> 0.15	✓
Minimum Element Quality	0.2685	> 0.10	✓
Maximum Aspect Ratio	7.7755	< 10	✓

Table 3.2 Summary of Mesh Quality

3.6 Numerical Simulation Setup

Numerical simulation of the benchmark plate-fin heat sink has been done through ANSYS Fluent 2026 R1. Once the mesh was imported for the calculation domain, all relevant materials, boundary conditions, and solution parameters were set based on the benchmark analysis by Singh et al. [9]. The analysis was done at steady-state to determine the fluid and thermal characteristics of the benchmark heat sink. Numerical simulation setup for benchmark model validation has been done prior to analysis of proposed SPF heat sink.

3.6.1 Material Properties

Properties of the materials utilized in the numerical simulation were provided in ANSYS Fluent prior to the governing equations solving. In this case, air was chosen as a working fluid, while aluminium was considered the material for heat sink and electronic chip. Material properties were considered constant during the whole process of the numerical simulation to ease the analysis and be consistent with the benchmarking approach methodology [9]. *In the present numerical study, aluminium was assigned as the material for both the heat sink and the chip during the CFD simulation in ANSYS Fluent. This simplification was adopted to maintain consistency in the numerical model used for the benchmark validation.*

Material	Density (kg/m ³)	Specific Heat (J/kg·K)	Thermal Conductivity (W/m·K)
Air	1.225	1006.43	0.0242
Aluminium (Heat Sink)	2719	871	202.4
Aluminium (Electronic Chip)	2719	871	202.4

Table 3.3 Material Properties Used in the Simulation

3.6.2 Boundary Conditions

Boundary conditions that matched the problem were applied to perform forced convection cooling simulation for the benchmark heat sink. Velocity inlet was defined at the inlet boundary of the simulation domain, whereas pressure outlet was used at the outlet. Heat source of the electronic chip was given according to the heat input mentioned in the benchmark case, and the surfaces were considered to be no-slip walls.

Parameter	Value
Inlet	Velocity Inlet
Outlet	Pressure Outlet
Heat Source	As per benchmark paper
Working Fluid	Air
Heat Sink Material	Aluminium
Wall Condition	No-slip

Table 3.4 Boundary Conditions

3.6.3 Solver Parameters

The problem was simulated using the pressure-based steady state solver provided by ANSYS Fluent. The energy equation was switched on for modelling heat transfer processes, and the RNG k- ϵ model with standard wall functions was used to model turbulent flow. Pressure-velocity coupling was realized using the SIMPLE approach, and second order discretization techniques were utilized [9].

Parameter	Selection
Solver Type	Pressure-Based
Time	Steady
Energy Equation	Enabled
Turbulence Model	RNG k- ϵ
Pressure-Velocity Coupling	SIMPLE
Spatial Discretization	Second Order

Table 3.5 Solver Settings

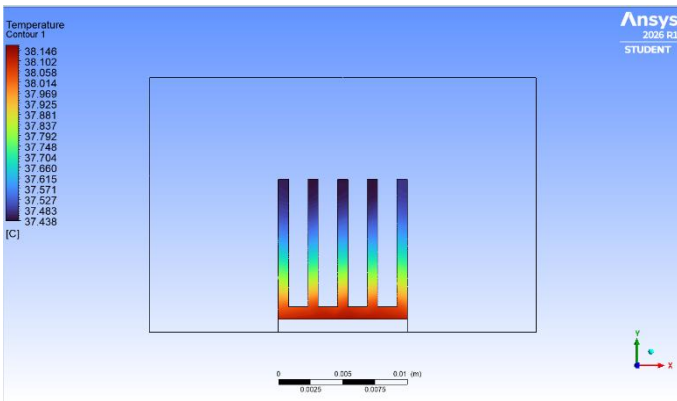
3.6.4 Solution Initialization and Convergence

Hybrid Initialization method was used to initialize the numerical solution. The run was stopped once the residual of continuity, momentum, energy, and turbulence equations converged as per the convergence criterion. Alongside checking for the residuals, the convergence of the maximum temperature and pressure drop values was also evaluated to make sure that the numerical solution has converged.

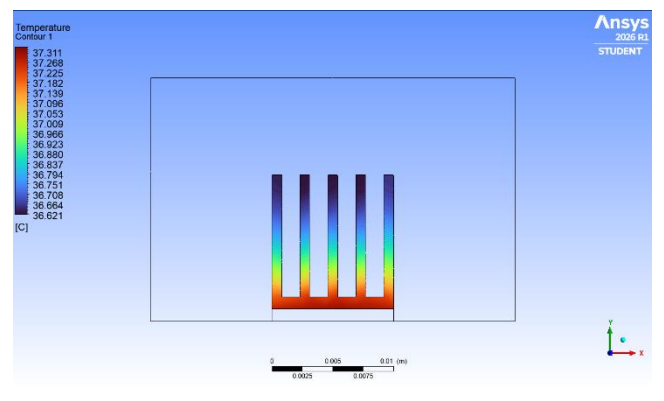
3.7 Benchmark Validation and Results

For verification of the developed numerical model, CFD analysis was conducted using six different operating conditions. Comparison was made between the temperature contours and chip average temperatures from the present analysis and the benchmark data available from Singh et al. [9]. The comparison reveals that the numerical method used in this work is quite accurate and reliable.

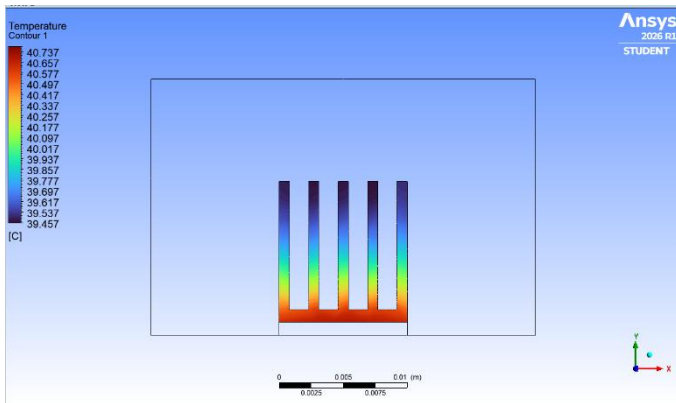
3.7.1 Temperature Contours



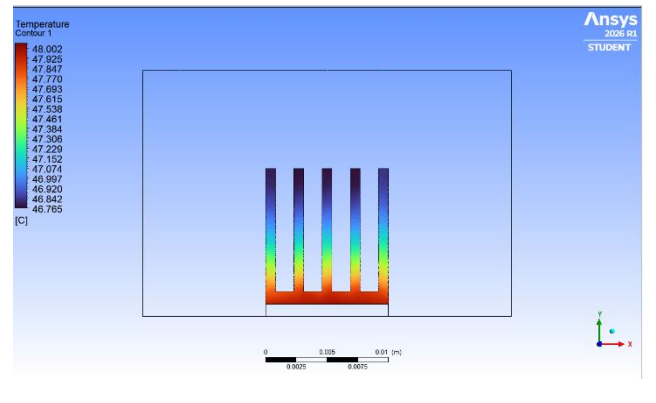
(a) Temperature contour at $V = 3$ m/s, $T_{\text{air}} = 15^\circ\text{C}$ and $q_{\text{gen}} = 1 \times 10^7$ W/m³.



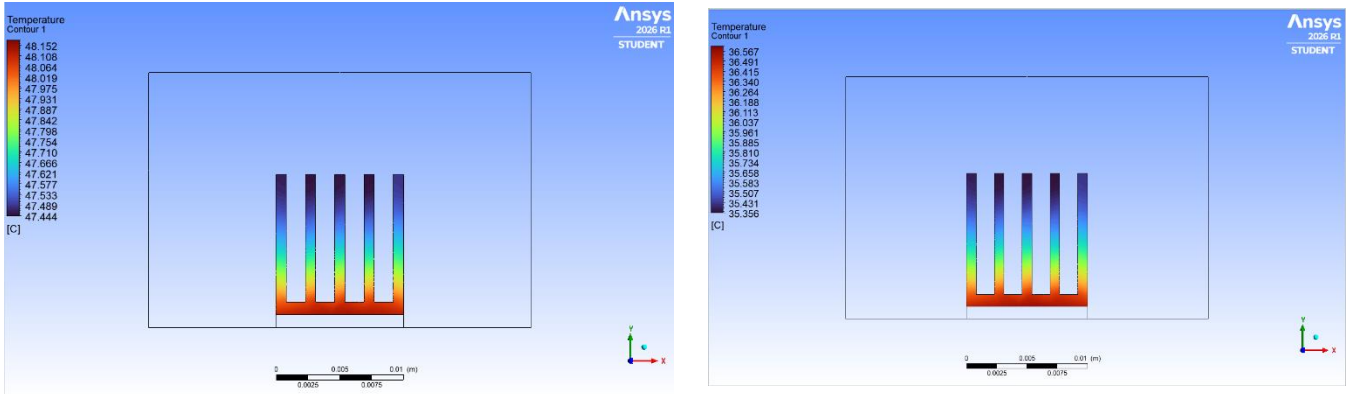
(b) Temperature contour at $V = 4$ m/s, $T_{\text{air}} = 25^\circ\text{C}$ and $q_{\text{gen}} = 1 \times 10^7$ W/m³.



(c) Temperature contour at $V = 2$ m/s, $T_{\text{air}} = 15^\circ\text{C}$ and $q_{\text{gen}} = 1.75 \times 10^7$ W/m³



(d) Temperature contour at $V = 3$ m/s, $T_{\text{air}} = 25^\circ\text{C}$ and $q_{\text{gen}} = 1.75 \times 10^7$ W/m³



(a) Temperature contour at $V= 3 \text{ m/s}$, $T_{\text{air}} = 35^\circ\text{C}$ and $q_{\text{gen}}= 1 \times 10^7 \text{ W/m}^3$.

(b) Temperature contour at $V= 4 \text{ m/s}$, $T_{\text{air}} = 15^\circ\text{C}$ and $q_{\text{gen}}= 1.75 \times 10^7 \text{ W/m}^3$.

Figure 3.9 Temperature contours for different cases for Benchmark Study

3.7.2 Comparison of Numerical Results with Benchmark Study

Case	Air Velocity (m/s)	Inlet Air Temperature ($^\circ\text{C}$)	Heat Generation (W/m^3)	Present Study (K)	Benchmark Study (K)	Error (%)
1	3	25	1.00×10^7	310.230	306.80	1.18
2	4	25	1.00×10^7	310.49	315.36	1.54
3	2	15	1.75×10^7	313.922	322.25	2.58
4	3	25	1.75×10^7	321.197	329.76	2.59
5	3	35	1.00×10^7	321.327	326.08	1.45
6	4	15	1.75×10^7	309.765	318.44	2.82

Table 3.6 Comparison of Present CFD Results with Benchmark Study

3.7.3 Discussion of Validation Results

Results obtained from the current CFD simulations were compared with the benchmark results given by Singh et al. [9] as shown in Table 3.6 below. From the table, it can be seen that there is good correlation between the current study and the benchmark results, for all the six different conditions tested. Percentage errors range from **1.18%** to **2.82%**, giving a total average validation error of **2.03%**.

Temperature contour plots obtained from the current simulation are also comparable with those given in the benchmark. Under all operating conditions, it can be seen that the temperature rises to its maximum values at the heat source and heat sink bases, but decreases along the height of the fins due to heat conduction through the fins and convective heat transfer to the surroundings. An increase in the air inlet velocity improves the cooling performance, hence the reduction in the average chip temperature; however, increasing heat generation increases the operating temperatures.

The small discrepancies between the current and benchmark results can be linked to the variations in the quality of mesh, numerical discretization approach, convergence criteria, and solver techniques. Given that the highest value of validation error is less than **3%** and the average validation error is just **2.03%**, the numerical model established can be said to have an excellent match with the benchmark case. Hence, the numerical approach used in the current research is deemed valid and is employed in the thermal analysis of the proposed **Staggered Perforated Fin (SPF)** heat sink in the next chapter [9].

CHAPTER-4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter describes the numerical analysis of the **proposed SPF heat sink using ANSYS Fluent 2026 R1 Student Version**. Having successfully validated the numerical model on the basis of the benchmark study in Chapter 3 [9], the same computational approach was used for the assessment of the thermal characteristics of the proposed SPF heat sink under the same operating conditions.

The proposed SPF heat sink was designed by making circular staggered perforations in the fins without changing any other dimensions of the fin, which is equal to the dimension of the fin of the benchmark heat sink [9]. The purpose of the perforations is to facilitate airflow through the fin passages, intensify fluid mixing and, thus, the rate of the convective heat transfer. As a consequence, the operating temperature of the electronic chip is supposed to be lower than that of the straight-fin heat sink.

To provide an unbiased comparison, the computational domain, the process of meshing, the material parameters, the boundary conditions and the settings of the numerical solver remained the same as in Chapter 3 [9] for the benchmark model. The only difference is the fin geometry.

Six different operation cases were used for the simulations with varying values of air velocity, air temperature, and volumetric heat generation rate. The thermal performance of the SPF heat sink has been assessed in terms of temperature profile, average chip temperature, and comparison with the conventional straight-fin heat sink. The results presented in this chapter offer an exhaustive evaluation of the effectiveness of the proposed SPF heat sink design.

4.2 Development of the Proposed Staggered Perforated Fin (SPF) Heat Sink

After the successful validation of the classical straight-fin heat sink as discussed in Chapter 3, a novel heat sink structure was designed by perforating circular holes in all the fins. The novel heat sink design is made up of five rectangular aluminium fins with circular holes, which are arranged in a staggered fashion while keeping the same dimensions as the benchmark heat sink design [9].

The aim of using perforations is to increase the air flow inside the fins to increase the convective heat transfer. Staggering of the holes will help in allowing the cooling air to

pass through the holes, which in turn would create turbulence within the fins and disturb the thermal boundary layer over the fins.

The selected perforation scheme of the current research includes five circular holes on each fin with a diameter of **0.80 mm**. There are two vertically-staggered rows of holes whose horizontal separation is **4.00 mm**, and the vertical separation between successive holes is **2.30 mm**. The first hole is located **1.50 mm** away from the upper boundary of the fin, and the centre line of the first row is **2.00 mm** away from the left edge of the fin. The dimensions of the perforation scheme are demonstrated in Figure 4.2.

Apart from introducing the staggered perforations, no changes were introduced into the other geometric parameters such as fin sizes, fin spacing, base plate dimensions, dimensions of the electronic chip, domain dimensions, materials, and operating conditions that are all same as the validated benchmark model [9]. Thus, any modification in the thermal performance of the heat sink is due to the modified fin design alone.

The three-dimensional SPF heat sink model was generated using SolidWorks and then imported into ANSYS Fluent 2026 R1 Student Version for analysis. The validated numerical approach presented in Chapter 3 was used to examine the thermal performance of the proposed heat sink at six different operating conditions.

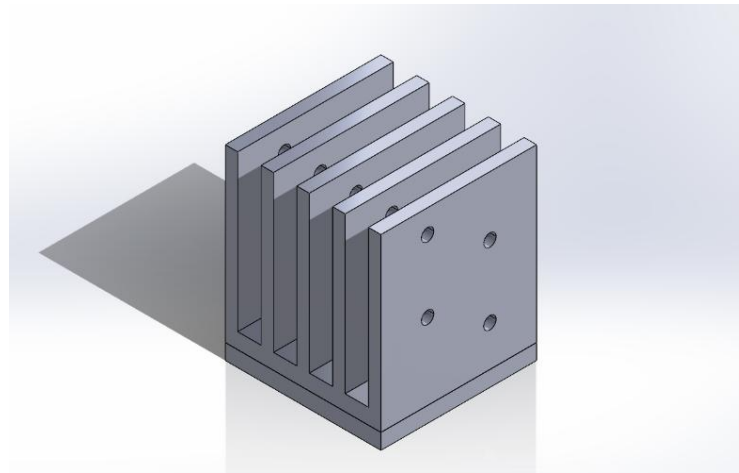


Figure 4.1: SolidWorks model of the proposed Staggered Perforated Fin (SPF) heat sink.

Parameter	Value
Fin type	Straight fin with staggered perforations
Number of fins	5
Holes per fin	5
Hole diameter	0.80 mm
Horizontal spacing between columns	4.00 mm
Vertical spacing between holes	2.30 mm
Distance of first hole from top edge	1.50 mm
Distance of first column from left edge	2.00 mm
Material	Aluminum

Table 4.1 Perforation Geometry of the Proposed SPF Heat Sink

4.3 Numerical Simulation of the Proposed SPF Heat Sink

Numerical simulation of the Staggered Perforated Fin heat sink (SPF) design was conducted with the use of ANSYS Fluent 2026. The validated numerical methodological approach, which was elaborated in Chapter 3, was applied to study the thermal characteristics of the redesigned heat sink. In order to conduct an unbiased comparison between the modified heat sink and the benchmark one, all simulation parameters were set to be the same as those, which were applied in the validation study [9].

The computational domain included the electronic chip, aluminium heat sink, and air enclosure around them. Three-dimensional conjugate heat transfer analysis was applied to simulate heat transfer processes within the solid bodies and forced convection within the air surroundings. Steady-state simulations based on the pressure-based solver were conducted.

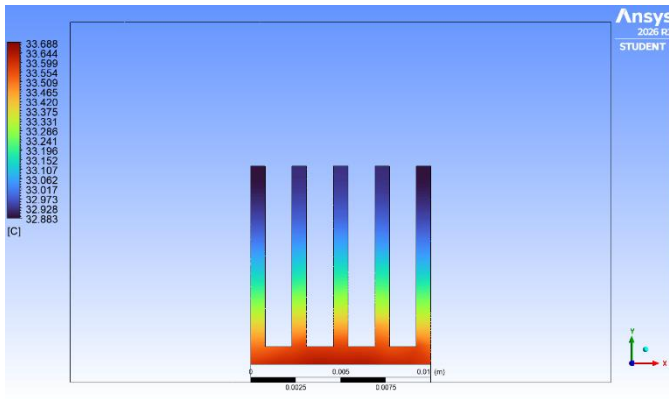
The working fluid in the simulations was air, and the material of the heat sink and the electronic chip was aluminium. RNG $k-\varepsilon$ turbulence model was applied to accurately predict turbulent flow through the fin passages and holes.

Thermal performance of the proposed SPF heat sink was carried out with six different operating conditions based on changes in the inlet air velocity, inlet air temperature, and volumetric heat generation rate. These six operating conditions were selected in order to have consistency in comparison with the previous research. Thus, it will be possible to compare the straight fin heat sink and the proposed SPF heat sink.

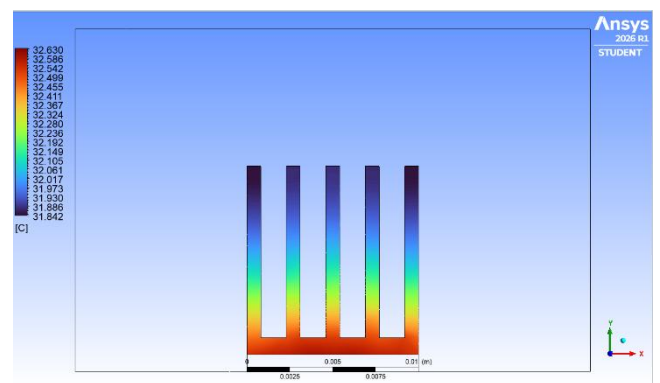
4.4 Temperature Distribution of the Proposed SPF Heat Sink

The temperature profile derived from the simulation results gives very important information regarding the thermal characteristics of the proposed SPF heat sink design. In order to assess the performance of the newly designed fin, the temperature contours of the SPF heat sink were compared with those of the conventional straight fin heat sink design based on six different operating conditions. The mean chip temperature obtained from each simulation run was considered as the basis for evaluating the performance of the heat sink.

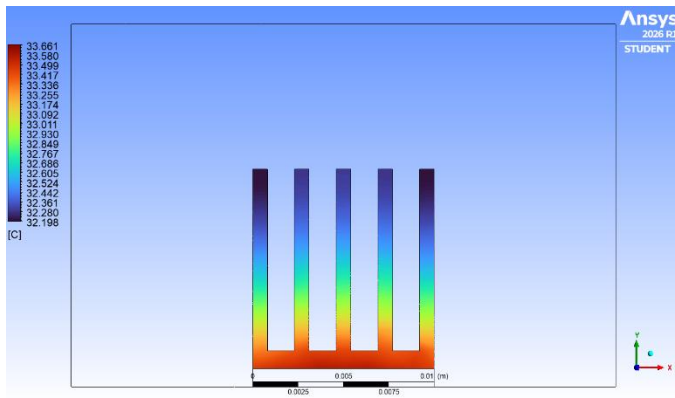
4.4.1 Temperature Contour



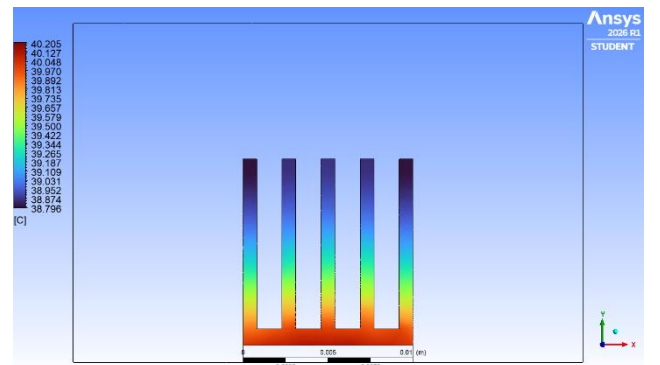
(a) Temperature contour at $V = 3 \text{ m/s}$, $T_{\text{air}} = 25^\circ\text{C}$ and $q_{\text{gen}} = 1 \times 10^7 \text{ W/m}^3$.



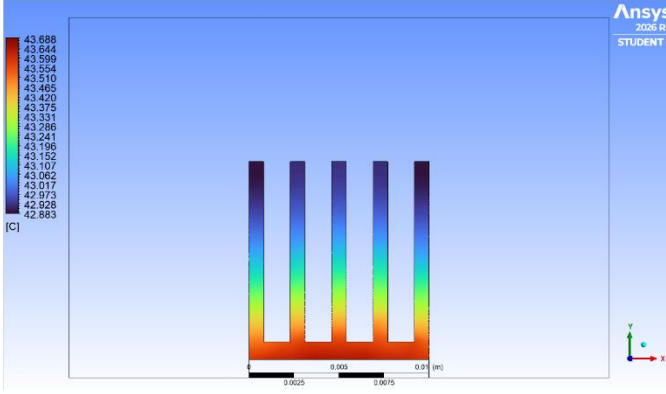
(b) Temperature contour at $V = 4 \text{ m/s}$, $T_{\text{air}} = 25^\circ\text{C}$ and $q_{\text{gen}} = 1 \times 10^7 \text{ W/m}^3$.



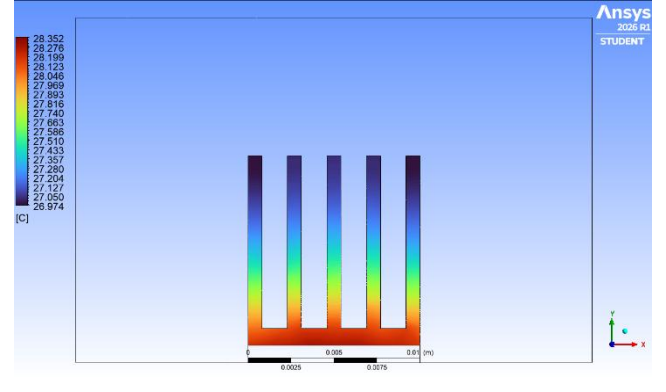
(c) Temperature contour at $V = 2 \text{ m/s}$, $T_{\text{air}} = 15^\circ\text{C}$ and $q_{\text{gen}} = 1.75 \times 10^7 \text{ W/m}^3$



(d) Temperature contour at $V = 3 \text{ m/s}$, $T_{\text{air}} = 25^\circ\text{C}$ and $q_{\text{gen}} = 1.75 \times 10^7 \text{ W/m}^3$



(a) Temperature contour at $V = 3 \text{ m/s}$, $T_{\text{air}} = 35^\circ\text{C}$ and $q_{\text{gen}} = 1 \times 10^7 \text{ W/m}^3$.



(b) Temperature contour at $V = 4 \text{ m/s}$, $T_{\text{air}} = 15^\circ\text{C}$ and $q_{\text{gen}} = 1.75 \times 10^7 \text{ W/m}^3$.

Figure 4.2 Temperature contours for different cases for Proposed SPF

4.5 Comparative Thermal Performance Analysis

The average temperatures of the chips have been compared in Table 4.2 for the present benchmark simulation, the benchmark analysis carried out by Singh et al. [9] and the Staggered Perforated Fin (SPF) heat sink using six different operating conditions. The validation of the benchmark simulation was carried out in the present study before analysing the proposed SPF heat sink using the same numerical technique.

The comparison between the average chip temperatures from the benchmark study [9] and the present numerical simulation of the conventional straight-fin heat sink and the proposed Staggered Perforated Fin (SPF) heat sink at six different operating conditions is illustrated in **Table 4.2**. The present numerical results for the conventional heat sink agree very well with the results of the benchmark study, validating the numerical procedure adopted in the current work. Following the validation process, the same numerical procedure was used to examine the thermal performance of the proposed SPF heat sink.

It can be seen that the average chip temperature of the proposed SPF heat sink is lower than that of the conventional heat sink at all six different operating conditions. It is because the use of staggered circular perforations in the fins increases the flow rate and hence causes more fluid mixing, disturbing the thermal boundary layer. As a result, heat transfer is improved.

Case	Air Velocity (m/s)	Inlet Air Temperature (°C)	Heat Generation (W/m ³)	Conventional Heat Sink (Present Study) (K)	Benchmark Study [9] (K)	Proposed SPF Heat Sink (K)
1	3	25	1.00×10^7	310.230	306.800	306.800
2	4	25	1.00×10^7	310.490	315.360	305.790
3	2	15	1.75×10^7	313.922	322.250	306.813
4	3	25	1.75×10^7	321.197	329.760	313.367
5	3	35	1.00×10^7	321.327	326.080	316.845
6	4	15	1.75×10^7	309.765	318.440	301.521

Table 4.2 Comparison of Average Chip Temperature

Case	Conventional Heat Sink (K)	Proposed SPF Heat Sink (K)	Temperature Reduction (K)	Percentage Improvement (%)
1	310.230	306.800	3.430	1.11
2	310.490	305.790	4.700	1.51
3	313.922	306.813	7.109	2.26
4	321.197	313.367	7.830	2.44
5	321.327	316.845	4.482	1.39
6	309.765	301.521	8.244	2.66
Average	314.655	308.523	5.966	1.90

Table 4.3 Improvement of Proposed SPF Heat Sink over the Conventional Heat Sink

Table 4.3 shows the extent to which the thermal performance of the SPF heat sink is improved over the thermal performance of the straight fin heat sink. In the cases studied, the thermal improvement is shown to reduce the temperatures in the range of 3.430 K to 8.244 K with an average improvement of 5.966 K equivalent to an average of about 1.90% improvement. The greatest thermal improvement is obtained in Case 6 and the least in Case 1.

This superior thermal performance of the heat sink is due to the presence of the staggered perforations in the form of circles that enable some portion of the cooling air to flow through the fins resulting in better air flow distribution, better fluid mixing, reduced thermal resistance and higher convective heat transfer coefficient.

4.6 Chapter Summary

In this chapter, numerical analysis of the proposed SPF heat sink in has been carried out using ANSYS Fluent 2026 R1 Student Version. In order to propose the SPF heat sink, the introduction of staggered circular perforations has been made on the conventional straight-fin heat sink with the same dimensions and operational parameters used in the benchmark model [9]. Six sets of simulation have been carried out to check the thermal efficiency of the SPF heat sink.

From the simulation results, it can be concluded that the SPF heat sink in consistently provided low average chip temperature compared to the conventional straight-fin heat sink at any set of operating condition. The addition of staggered perforations facilitated better air movement through the fin passages, fluid mixing and thus heat transfer due to convection. As a result, the average chip temperature dropped from **3.430 K to 8.244 K** depending on the operating conditions considered in this study. Thus, the average drop in average chip temperature was about **5.97 K** and an average improvement in thermal efficiency was **1.90%**.

The results prove that the designed SPF heat sink is better than the traditional heat sink in terms of performance and that it could be utilized to enhance heat removal in electronic cooling systems. The numerical simulation technique validated and the results obtained in this chapter form the basis for the conclusions made in the next chapter.

CHAPTER 5

CONCLUSION AND RECOMMENDATION FOR FUTURE WORK

5.1 Conclusion

This study aimed at analysing the thermal performance of Staggered Perforated Fin (SPF) heat sink used in the cooling of electronics. In the first instance, the conventional straight-fin heat sink was designed and validated using benchmark data of Singh et al. [9]. The results of the validation were in close agreement with the benchmark results, indicating the efficiency of the adopted computational methods.

From the validated model, another design of heat sink with staggered circular perforations was created and analysed under six distinct operating conditions through ANSYS Fluent 2026 R1 Student Version. The computational domain, physical parameters, meshing, boundary conditions, and other numerical methods remained the same as those in the benchmark model.

From the simulation results, it can be seen that the average chip temperature of the SPF heat sink was lower than that of the conventional straight-fin heat sink in all the operating conditions. The temperature difference varied between **3.430 K** to **8.244 K**, with an average difference of about **5.97 K** or **1.90%**.

On the whole, it can be concluded that the designed SPF heat sink is quite a good modification which allows one to enhance the heat dissipation capacity of the heat sink while retaining its overall geometrical design. The results show that staggered perforations can considerably increase the efficiency of plate-fin heat sinks.

5.2 Practical Applications

The new SP heat sink showed better thermal performance compared to the traditional straight fin heat sink since it lowered the average temperature of the chip during all the six operational scenarios. This enhanced cooling effect realized from the use of the staggered perforation design makes the proposed heat sink applicable in many practical applications.

Possible applications of the new SPF heat sink could be:

- a. **Cooling of computer CPUs and GPUs** that need effective cooling solutions.
- b. **Heating of IGBTs, MOSFETs, converters, and inverters** that are power electronic device.
- c. **Cooling of electronic control units and battery management systems** of electric vehicles.
- d. **Cooling of telecommunication and networking equipment** since their continuous operation leads to the heating of their components.

- e. **LED lighting systems** because they require an effective passive cooling solution.
- f. **Automated industrial controllers** and embedded electronic systems.
- g. Electronic medical equipment that needs an efficient temperature control system.
- h. Gaming consoles and other consumer electronic devices including **mini-PCs and high-end embedded electronics**.

The proposed SPF-based heatsink could be made by traditional methods of aluminium machining or through CNC drilling process, which would make it industrially feasible without any drastic changes to the existing process of manufacturing.

5.3 Recommendations for Future Work

Despite the fact that the proposed SPF heat sink showed better thermal efficiency, there is still room left for improvements and future studies. The following suggestions could be considered for future studies:

- a. Study the effects of different **hole diameters, number of holes, and different hole placement in terms of achieving optimal design for highest heat transfer**.
- b. Determine the influence of other perforations shapes (**elliptical, square, hexagonal, slots etc.**) on heat dissipation.
- c. Study the effects of using different heat sink materials (**copper, aluminium alloy, graphite, composite materials**).
- d. Validate experimental results of the proposed SPF heat sink by designing a prototype of the heat sink.
- e. Study the heat dissipation performance in transient working conditions and with different heat loads.
- f. Study the application of nanofluids and other hybrid cooling technologies along with the proposed SPF heat sink.
- g. Use various methods of optimizations, such as genetic algorithms or machine learning methods.
- h. Apply the proposed concept of heat sink to **higher power electronic devices and systems, data centres, electric vehicle power electronics, renewable energy converters, aerospace electronic equipment and others**.

CHAPTER 6

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